Question 1. Estimate the error if $P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$ is used to estimate the value of $f(x) = \cancel{\sin x}\ e^x$ ~~by~~ fill in the blanks below.

1. By Taylor's theorem, for each x, the error is

$$R_3(x) = f(x) - P_3(x) = \frac{f^{(4)}(c)}{4!}x^4 = \underline{\quad \frac{e^c}{24} \quad} x^4$$

for some c in between 0 and x.

2. Noting that $|c| \le |x|$, the error is bounded by (fill in a number)

$$|R_3(x)| \le \underline{\ \tfrac{1}{24}\ }\, e^{|x|}|x|^4.$$

3. If we were to consider all x in $(-1, 1)$, an upper bound for the error is (fill in a number)

$$|R_3(x)| \le \underline{\ \tfrac{1}{8}\ } \quad \text{using } e^{|x|} \le 3 \quad |x|^4 \le 1$$

(or $\frac{e}{24}$)

4. In general, if we were to consider all x in $(-a, a)$, an upper bound for the error is

$$|R_3(x)| \le \underline{\quad \tfrac{1}{24} e^a \cancel{x} a^4 \quad}$$

depending on the value of a.

If we want to find an interval $\left(-\frac{1}{N}, \frac{1}{N}\right)$ such that the error is less than 0.001, we need to solve the inequality

$$|R_3(x)| \le \underline{\quad \frac{e^{1/N}}{24}\,\frac{1}{N^4}\ \left(\le \frac{1}{9N^4}\right) \quad} < 0.001,$$

so one possible value of N is $\underline{\ 10\ }$. (feel free to be wasteful!)

Question 2. Calculate an upper bound for the error if the following $P_n(x)$ is used to estimate $f(x)$ on $(-0.1, 0.1)$. Moreover, find an interval $(-a, a)$ such that the error is less than 0.001.

a) $P_1(x) = 1 + \frac{x}{2}$ for $f(x) = \sqrt{1+x}$

$f'(x) = \frac{1}{2(1+x)^{1/2}} \qquad f''(x) = -\frac{1}{4(1+x)^{3/2}}$

$|R_1(x)| \le \frac{1}{8(1+c)^{3/2}}|x|^2$

For c and x in $(-0.1, 0.1)$, $|R_1(x)| \le \frac{0.01}{4} = 0.0025$ using $\frac{1}{(1+c)^{3/2}} \le 2$.

We can use $a = 0.01$, then we still have $\frac{1}{(1+c)^{3/2}} \le 2$,

so $|R_1(x)| \le \frac{0.01^2}{4} \le 0.001$

b) $P_4(x) = x - \frac{x^3}{6}$ for $f(x) = \sin(x)$

(this P_4 is indeed a fourth-degree approximation of $\sin(x)$)

$f^{(5)}(x) = \cos x$

$|R_4(x)| \le \cancel{\frac{|\cos c|}{120}|x|^5}\ \frac{|\cos c|}{120}|x|^5 \le \frac{|x|^5}{120}$

For x ~~and~~ in $(-0.1, 0.1)$, $|R_4(x)| \le \frac{0.1^5}{120}$

Because $\frac{0.1^5}{120} < 0.001$, $a = 0.1$ already works.

c) $P_2(x) = x + x^2$ for $f(x) = \sin(x)e^x$

$f^{(3)}(x) = (\sin(x) + 3\cos(x) - 3\sin(x) - \cos x)e^x$

$= (-2\sin(x) + 2\cos(x))e^x$

$|R_2(x)| \le \frac{4e^{|x|}}{6}|x|^3$

For x in $(-0.1, 0.1)$, $|R_2(x)| \le 2 \cdot 0.1^3$

$= 0.002$

Using $e^{|x|} \le 3$.

We can use $a = 0.01$,

then we still have $e^{|x|} \le 3$,

so $|R_2(x)| \le 2 \cdot 0.01^3 \le 0.001$.

d) $P_3(x) = \frac{x^3}{3}$ for $f(x) = \int_0^x \sin(t^2)\,dt$

Note $f(x) = \int_0^x \sum_{n=0}^{\infty} \frac{(-1)^n t^{4n+2}}{(2n+1)!}\,dt = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+3}}{(4n+3)(2n+1)!}$

For all x, this is alternating, so

$|R_3(x)| = \left|\sum_{n=1}^{\infty} \frac{(-1)^n x^{4n+3}}{(4n+3)(2n+1)!}\right| \le \frac{|x|^7}{42}$

when x is in $(-0.1, 0.1)$, $|R_3(x)| \le \frac{0.1^7}{42}$

We can just use $a = 0.1$ because $\frac{0.1^7}{42} < 0.001$